

EASTERN INTERNATIONAL UNIVERSITY
BECAMEX BUSINESS SCHOOL



Course: MIS 451 - Machine Learning for Business

Topic: E-commerce Shipping via Classification Analysis

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1. Business Context and Objectives

1.1 Business Context

In today's competitive business environment, organizations increasingly rely on data-driven decision-making to improve operational efficiency, reduce costs, and enhance customer satisfaction. Machine Learning (ML) techniques allow businesses to identify patterns from large datasets and make accurate predictions that support strategic planning and operational management.

This project focuses on developing a classification model capable of predicting whether an order is able to reach its destination on time using prior data. The purpose of the project is not only to build predictive models, but also to understand the relationships between variables, evaluate different machine learning approaches, and translate technical findings into optimizing the operation and overseeing bottlenecks.

The dataset used in this project contains almost 11,000 observations and multiple independent variables, satisfying the requirements of the assignment. Through data preprocessing, exploratory analysis, model development, and evaluation, the project demonstrates a complete machine learning workflow for business analytics applications.

1.2 Project Objectives

The project aims for three core objectives:

1. Conduct comprehensive data analysis to explore the structure, distribution, and relationships between variables to identify factors that influence delivery status.
2. Develop and compare models implementing four classification algorithms, including three traditional Machine Learning models (KNN, SVM, Logistic Regression) and one Deep Learning model (ANN), and then select the best model based on experimental results.
3. Transform the results into business recommendations, translating technical data into concrete actions that the operations department can implement immediately.

2. Data Collection and Characteristics

2.1 Dataset Description

The dataset, obtained from Kaggle and called the E-Commerce Shipping Dataset, simulates the logistics data of an e-commerce business. The original data consists of 10,999 observations and 12 variables. After removing two columns with no analytical value, namely ID (order identifier) and Gender (which had no significant correlation with delivery status, otherwise would cause gender inequality issues), the working dataset contains 10 variables, including 6 numeric variables, 3 categorical variables, and 1 binary target variable.

Variables	Data Type	Description
Warehouse_block	Categorical	Warehouse Area (A, B, C, D, F)
Mode_of_Shipment	Categorical	Shipping Method (Ship, Flight, Road)
Customer_care_calls	Numerical	Number of times customers called customer support
Customer_rating	Numerical	Customer rating (1–5 scale)
Cost_of_the_Product	Numerical	Product value (USD)
Prior_purchases	Numerical	Number of previous purchases by the customer
Product_importance	Categorical	Priority level (Low, Medium, High)
Discount_offered	Numerical	Discount applied (%)
Weight_in_gms	Numerical	Product weight (grams)
Reached.on.Time_Y.N	Target (Binary)	0 = On-time delivery 1 = Late delivery

Table 1. Description of variables in the E-Commerce Shipping Dataset

3. Exploratory Data Analysis (EDA)

3.1 Dataset Overview

After loading the data, the team performed a comprehensive check of data structure and quality before proceeding to detailed analysis.

Target variable distribution: In the training set, class 1 (late delivery) accounted for 6,567 observations (59.7%) and class 0 (on-time delivery) accounted for 4,433 observations (40.3%). This slight imbalance is the reason the team chose F1-macro instead of accuracy as the primary evaluation metric.

3.2 Train-Test Split

Data splitting is performed early, before all transformation steps, to ensure the independence of the test set. Parameters used: `test_size=0.2`, `stratify` by target variable, `random_state=99`. This is a crucial technical decision: any information from the test set (including mean and distribution) must not influence the preprocessing on the training set. So in the end, we have: The training set consisted of 8,799 observations, and the test set consisted of 2,200 observations.

3.3 Data Cleaning

The test results showed that the dataset was of good quality from the start: the `df.duplicated().sum()` code returned 0, indicating there were no duplicate rows. The `df.isnull().sum()` code applied to both training and testing returned 0 on all columns without missing values. This allowed the team to skip the input step and focus directly on analysis.

As confirmed in the Data Cleaning step, the dataset has no missing values in either the train or test set. Therefore, the imputation step is not necessary. This is a favorable condition, allowing the team to focus entirely on feature engineering and model development.

3.4 Exploratory Data Analysis (After Cleaning)

3.4.1 Univariate Analysis

There are 4 categorical columns included "Warehouse_block", "Mode_of_Shipment", "Product_importance", and "Reach.on.Time_Y.N". Through the countplot and pie chart for the categorical variable, our group found that warehouse F accounted for the largest proportion, indicating uneven processing capacity among warehouses. Regarding the mode of shipment, sea transport (Ship) completely dominated with approximately 68%, posing a high risk of delays due to long transit times. Meanwhile, the product importance level ("Product_importance") had three levels: Low, Medium, and High, where around 91% of it was distributed quite evenly on the first 2 levels, where as only around 9% was in the level "High". Notably, the target variable recorded approximately 60% of orders being delivered late (class 1), this slight imbalance requires careful consideration when selecting metrics to evaluate the model.

For the numerical variables, the histogram and KDE curve reveal very characteristic distribution patterns. The discount variable ('Discount_offered') shows a significant rightward bias, with the majority of orders receiving discounts of 0 to 30%, but an outlier group exhibiting extremely high discounts (larger than 50%). The number of calls ('Customer_care_calls') is concentrated between 3 to 5 calls, indirectly reflecting problematic processing that requires customers to contact the company multiple times. Finally, the product volume ('Weight_in_gms') is fairly uniform, ranging from 1,000 to 6,000 grams, while the cost of goods sold ('Cost_of_the_Product') is nearly symmetrical, ranging from \$150 to \$300, characteristic of mass market goods.

3.4.2 Bivariate Analysis

a) Categorical Features vs Target (Countplot of 2 variables by target)

Mode_of_Shipment: Sea transport (Ship) is the most frequently used method, far surpassing Air (Flight) and Road (Road). Product_importance: The majority of items are classified as 'low' and 'medium'. However, in all three methods of both features, the number of delayed orders (1) is always significantly higher than the number of on-time orders (0). The

delay/on-time ratio is fairly uniform across methods, indicating that this variable, standing alone, may not provide much classification power.

b) Numerical Features vs Target (KDE plot of 6 variables by target)

Based on the Density Plot, we can divide variables into two groups based on their target:

- Group with weak classification power (High overlap distribution):

Customer_care_calls & Customer_rating & Prior_purchases: The distribution density of on-time and late deliveries is almost identical. This indicates that the number of customer care calls, ratings, or previous purchases are not clear indicators of late deliveries.

Cost_of_the_Product: The graph shows fluctuations, but overall, the cost distribution of on-time and late deliveries remains highly intertwined.

- Group with strong classification power (Distribution with separation):

Discount_offered: This is the most important variable.

- On-time delivery (0) is concentrated in an extremely sharp peak at low discount levels (from 0 to about 10%). Conversely, late delivery (1) has a tail extending to the 60-70% discount level. It can be concluded that most products with large discounts (over 10%) are delivered late.

Weight_in_gms: The graph shows a bimodal shape.

- The late delivery group (1) is very densely concentrated in the light goods segment (about 1000g - 3000g). The on-time delivery group (0) forms a large peak in the heavy goods segment (about 4000g - 6000g). The range between 3000g and 4000g is almost a "low point" with little data available.

Scatter/Strip Plots

- Discount Offered Difference: The chart shows the average discount offered on late orders is 18.5, more than three times higher than the average discount offered on on-time orders of 6. The blue data block (Late) covers the range from 10 to 65, while the red dot (On Time) is just below 10.
- Weight Difference (Weight_in_gms): The average weight of on-time orders (4200g) is considerably heavier than that of delivered orders (3300g). This analysis tip notes that heavier orders tend to perform better on-time.

Correlation Heatmap (after encoding): No pairs of variables have an absolute correlation larger than 0.8 so no serious multicollinearity. Discount_offered and Weight_in_gms have the clearest positive correlation with the target variable.

3.4.3 Outlier Exploration

IQR testing revealed that the variables 'Prior_purchases' and 'Discount_offered' exhibited a large number of outliers, 797 and 1,737 respectively. However, the research team decided to retain all these outliers because they accurately reflect business reality representing loyal customers, wholesale buyers, or genuine flash sales and clearance events, rather than data errors.

4. Data Transformation

4.1 Feature Encoding and Scaling

One-Hot Encoding

Three categorical variables: Warehouse_block, Mode_of_Shipment, and Product_importance. The variables are encoded using scikit-learn's OneHotEncoder. The encoder is fitted on the training set and only transformed (not fitted) on the test set, adhering to the principle of preventing data leakage. This process creates 11 new binary columns to replace the original 3 columns:

- Warehouse_block → 5 columns (A, B, C, D, F)
- Mode_of_Shipment → 3 columns (Flight, Road, Ship)
- Product_importance → 3 columns (High, Low, Medium)

Preventing Data Leakage

- For the training set: Use the '.fit_transform()' method to "learn" the categories and perform the transformation.
- For the test set: Only use the '.transform()' method based on what has been learned from the training set. This is crucial, as it ensures the test set remains entirely new data, preventing data leakage from the test set into the training process.

Feature Space Update

After successfully creating the One-Hot columns, the original categorical columns in string format were removed (dropped) from both the Train and Test datasets to avoid information redundancy. The entire dataset was now fully digitized.

Feature and Target Variable Separation

- Training Set: Xtrain: 8,799 samples (rows) with 17 features (columns). ytrain: 8,799 corresponding target labels.
- Testing Set: Xtest: 2,200 samples with 17 features. ytest: 2,200 target labels.

Standard Scaling

The entire feature matrix is normalized to a distribution with mean equals 0 and standard deviation equals 1 using StandardScaler. The scaler is fitted_transformed on the Xtrain and only transformed on the Xtest.

5. Model Development (Classification)

5.1 Evaluation Methodology

The entire model comparison process was performed using K-Fold Cross-Validation with $k = 10$ on the training set. The primary metric was the F1-macro, the unweighted average of the F1-scores of the two classes. The reasons for choosing this metric were: accuracy can be misleading when the classes are not perfectly balanced (60/40); the F1-macro ensures that both classes are evaluated fairly, accurately reflecting the model's true capabilities in a business context where both types of prediction errors (missing late orders and incorrectly reporting on-time orders) have cost consequences.

5.2 Model Selection Rationale

The group selected four models representing different machine learning philosophies, creating a comprehensive comparative picture:

- KNN (K-Nearest Neighbors): A non-parametric algorithm that predicts based on the k nearest points in the feature space. It represents an instance-based learning approach, heavily reliant on the quality of data normalization. GridSearchCV was applied to find the optimal k .
- SVM (Support Vector Machine): A hyperplane search with maximum margin separation. Effective in multidimensional spaces, suitable for well-scaled data.
- Logistic Regression: A fundamental linear model for binary classification. It serves as a baseline to assess whether more complex models truly deliver significant improvements.
- ANN (Artificial Neural Network - Deep Learning): A multi-layer neural network with the architecture Dense(64, ReLU) $\rightarrow$ Dropout(0.3) $\rightarrow$ Dense(32, ReLU) $\rightarrow$ Dropout(0.2) $\rightarrow$ Dense(1, Sigmoid), optimized using Adam ($lr=0.001$) and binary cross-entropy loss. EarlyStopping is applied to avoid overfitting.

6. Model Evaluation and Comparison

6.1 Model Comparison and Selection

After testing and evaluating the performance of various algorithms, the test results revealed a significant difference in predictive ability between the models. Logistic Regression ranked last with an F1-macro CV of 0.6222, indicating that the linear boundary was not complex enough for this data; in particular, the F1 class 0 was only 0.5607, suggesting the model struggled with accurate single-time identification. KNN achieved an F1-macro CV of 0.6342, but the large gap between training accuracy (72.72%) and the CV score suggested a slight overfit. SVM achieved the highest F1-macro CV among traditional ML (0.6468) thanks to its RBF kernel learning non-linear boundaries. ANN was chosen as the final model: its macro average had the highest F1 training ($0.7044 > \text{SVM } 0.6998$), and its architecture with Dropout and EarlyStopping was better regularized than KNN, making it suitable for practical implementation.

6.2 Quantitative Results

When evaluated overall across the test set, the artificial neural network (ANN) model achieved a fairly stable accuracy of 67.41% for all 2,200 test samples. However, for a more in-depth understanding of its ability to classify specific details for on-time and delayed orders, the results are shown in the table below:

Target class	Precision	Recall	F1-Score	Support
0 (Delivered on time)	0.5630	0.8568	0.6795	887
1(Delivered late)	0.8506	0.5506	0.6685	1313
Accuracy			67.41%	2200
Macro average	0.7068	0.7037	0.6740	2200
Weighted average	0.7346	0.6741	0.6729	2200

7. Interpretation and Business Insights

7.1 Strength: High Cost-Efficiency

The core strength of the ANN model is its 85.1% accuracy for the 'Delayed Delivery' class. In actual operation, out of every 100 alerts the model generates, 85 are accurate. Only 15 are false alarms. This is a sufficiently reliable threshold to deploy an automated alerting system without overwhelming the processing team. From 2,200 test orders, the model generated 850 alerts (723 TP + 127 FP), successfully preventing 723 delayed delivery cases, equivalent to 32.9% of the total test orders.

7.2 Limitation: Coverage Volume

Recall class 1 only reached 55.06%, meaning 590 out of 1,313 delayed orders were missed. Customers who receive late orders without prior notification will react much more negatively than those who are proactively notified. Two technical improvements: (1) Lower the classification threshold from 0.5 to 0.35 – 0.40 to increase Recall, accepting an increase in FP, adjusting based on the FP/FN cost trade-off in each context; (2) Add new features: time of year, delivery address, F warehouse inventory status.

7.3 Strategic Recommendation

Based on data analysis and model results, the team proposes three specific courses of action for management:

- Implement a real-time alert system: Integrate an ANN model into the order management system (OMS). Each newly created order will be automatically assessed for risk, and those with a high probability of delay (>threshold) will be prioritized in

the processing queue. This allows the logistics team to proactively allocate resources instead of reacting after an incident.

- Manage promotional campaigns in line with logistics capacity: EDA analysis shows that Discount_offered is one of the factors most strongly associated with the rate of delayed deliveries. This suggests that flash sales or large-scale promotions need to be planned in advance with coordination between the Marketing and Logistics departments, ensuring that warehouse and transportation capacity are expanded accordingly before the campaign begins.
- Improving customer care service quality: High customer care calls correlate with high delivery delay rates, this is not just a consequence but may be an early warning sign of a problem. The customer service department can be equipped with tracking tools to prioritize orders at risk, improve NPS, and reduce cancellation rates.

In the long term, the team recommends conducting periodic (quarterly) model retraining to keep pace with changes in customer behavior and the delivery network, and expanding the feature set with geographic and time-sensitive data to improve coverage (Recall).

8. GitHub Repository Links

Pham Thuy Huyen: <https://github.com/huyentpham/MIS-451>

Vo Thi Hoai Anh: <https://github.com/HoaiAnh2/MIS-451>

Le Anh Khuong: <https://github.com/Khuongeez/MIS-451>